



Operations Research

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Solution to the Training Exercises, Sheet 10

Training Exercises

Exercise T10.1. *Gradient Descent vs. Newton's Method*

Problem

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x_1, x_2) = x_1^2 + x_1x_2 + 1.5x_2^2 + x_1 - 4x_2 + 3$.

- (a) Compute the first iteration $x^{(1)}$ of the Gradient Descent method starting from $x^{(0)} = (1, 1)^T$ with a constant step size $\eta = 0.5$.
- (b) Compute the first iteration $x^{(1)}$ of Newton's Method starting from the same initial point $x^{(0)} = (1, 1)^T$.
- (c) Discuss the behavior of Newton's Method on this function compared to Gradient Descent.

Solution

- (a) First, compute the gradient:

$$\nabla f(x) = \begin{pmatrix} 2x_1 + x_2 + 1 \\ x_1 + 3x_2 - 4 \end{pmatrix}$$

At $x^{(0)} = (1, 1)^T$, $\nabla f(1, 1) = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$. The Gradient Descent step is:

$$x^{(1)} = x^{(0)} - \eta \nabla f(x^{(0)}) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.5 \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

- (b) The Hessian matrix is constant:

$$H_f = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

The inverse Hessian is $H_f^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$. The Newton step is:

$$x^{(1)} = x^{(0)} - H_f^{-1} \nabla f(x^{(0)}) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$x^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2.4 \\ -0.8 \end{pmatrix} = \begin{pmatrix} -1.4 \\ 1.8 \end{pmatrix}$$

- (c) Because the objective function f is a strictly convex quadratic function, its second-order Taylor expansion is globally exact. Therefore, Newton's Method converges to the exact global minimum $x^* = (-1.4, 1.8)^T$ in a single iteration. Gradient Descent, utilizing only linear approximations, generally requires many iterations to converge.

Exercise T10.2. Lagrange multipliers

Problem

Consider the following function and the corresponding constraint:

$$f(x, y) = -e^{x^2+y^2} \quad (1)$$

$$s.d. \quad g(x, y) = 2x + y = 5 \quad (2)$$

- (a) Show that f is concave.
- (b) Set up the corresponding Lagrange function $L(x, y, \lambda)$ for the restricted problem.
- (c) Determine the critical points (x^*, y^*, λ^*) of the Lagrange function. Decide and explain whether (x^*, y^*) is a saddle, maximum, or minimum point of the function f .
- (d) Let (x', y', λ') be a point such that

$$\left(\frac{\partial L}{\partial \lambda} \right) (x', y', \lambda') = 0.$$

What can we then say about $g(x', y')$?

Solution

- (a) From the lecture we know: f is concave $\Leftrightarrow -f$ is convex. We also know that if a function h is twice continuously differentiable and its Hessian matrix Hh is everywhere positive semi-definite, then h is convex. Since f is twice continuously differentiable, it is enough to show that $H(-f)$ is everywhere positive semi-definite. The Hessian matrix is calculated as follows:

$$H(-f)(x, y) = \begin{pmatrix} 2e^{x^2+y^2} + 4x^2e^{x^2+y^2} & 4xye^{x^2+y^2} \\ 4xye^{x^2+y^2} & 2e^{x^2+y^2} + 4y^2e^{x^2+y^2} \end{pmatrix} \quad (3)$$

We use the principal minor criterion to show that $H(-f)$ is positive definite. To do this, we observe that for all $x, y \in \mathbb{R}$:

$$2e^{x^2+y^2} + 4x^2e^{x^2+y^2} > 0 \quad (4)$$

$$\det(H(-f)) = \left(e^{x^2+y^2} \right)^2 \cdot [(2 + 4x^2) \cdot (2 + 4y^2) - 16x^2y^2] = 4e^{2(x^2+y^2)}(1 + 2x^2 + 2y^2) > 0 \quad (5)$$

Thus, $H(-f)$ is positive definite everywhere, which implies $-f$ is strictly convex, and hence f is strictly concave.

(b) The Lagrange function is given by:

$$L(x, y, \lambda) = f(x, y) + \lambda \cdot (5 - g(x, y)) \quad (6)$$

$$= -e^{x^2+y^2} + \lambda(5 - 2x - y) \quad (7)$$

(c) The critical points are given by the roots of the gradient of the Lagrange function:

$$\nabla L(x, y, \lambda) = \begin{pmatrix} -2xe^{x^2+y^2} - 2\lambda \\ -2ye^{x^2+y^2} - \lambda \\ 5 - 2x - y \end{pmatrix}. \quad (8)$$

Setting $\nabla L(x, y, \lambda) = 0$ yields the following system of equations:

$$\begin{aligned} \lambda &= -xe^{x^2+y^2} \\ \lambda &= -2ye^{x^2+y^2} \\ \Rightarrow -xe^{x^2+y^2} &= -2ye^{x^2+y^2} \Leftrightarrow x = 2y \\ 5 - 4y - y &= 0 \Leftrightarrow y = 1 \end{aligned}$$

It follows that the Lagrange function has exactly one critical point, given by $(x^*, y^*, \lambda^*) = (2, 1, -2e^5)$. Since f is globally concave and the constraint function $g(x, y) = 2x + y$ is affine linear, the unique critical point (x^*, y^*) is a global maximum.

(d) It holds true that:

$$\frac{\partial L}{\partial \lambda}(x, y, \lambda) = 5 - 2x - y = 5 - g(x, y). \quad (9)$$

Setting this derivative to zero yields $5 - g(x, y) = 0 \Leftrightarrow g(x, y) = 5$. This implies that the critical points (x', y') of the Lagrangian are precisely those that satisfy the primal constraint.

Exercise T10.3. Non-linear optimization - KKT conditions

Problem

Consider the following non-linear program.

Use without proof that a regularity conditions holds, i.e., the KKT conditions are necessary.

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|---|---|
| $\begin{aligned} (P) \quad \max \quad & \sin(x_1) + \cos(x_2) \\ & x_1 + x_2 \leq 2 \\ & 2x_1 - x_2 \geq 1 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}$ | <p>a) State the KKT conditions for (P). How do the KKT conditions change if we minimize instead of maximize?</p> <p>b) Using a), show that the point $x = (1, 1)$ is not a local maximizer for P.</p> <p>c) Consider the multiplier $\mu = 0$. Using a), derive the corresponding point x and check whether x is a local maximizer for (P). If so, is x also a global optimum and is it unique?</p> <p>d) Bonus (hard): Show that $x = (1, 1)$ is a local minimizer.</p> |
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Solution

Let $x = (x_1, x_2)$. Note that the constraint $x_1 \geq 0$ is already implied by the other constraints. The feasible region is a triangle with corners $C_1 = (0.5, 0)$, $C_2 = (2, 0)$, $C_3 = (1, 1)$.

a) We set

$$\begin{aligned} f(x) &= \sin(x_1) + \cos(x_2) \\ h_1(x) &= x_1 + x_2 - 2 \\ h_2(x) &= -2x_1 + x_2 + 1 \\ h_3(x) &= -x_2. \end{aligned}$$

Thus, we can write the optimization problem in the form familiar from the lecture as

$$(P) \quad \begin{aligned} \max \quad & f(x) \\ & h_1(x) \leq 0 \\ & h_2(x) \leq 0 \\ & h_3(x) \leq 0. \end{aligned}$$

The KKT conditions can be written as

$$\nabla f(x) - \sum_{i \in \{1,2,3\}} \mu_i \nabla h_i(x) = \underbrace{\begin{pmatrix} \cos(x_1) \\ -\sin(x_2) \end{pmatrix}}_{\nabla f} - \mu_1 \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\nabla h_1} - \mu_2 \underbrace{\begin{pmatrix} -2 \\ 1 \end{pmatrix}}_{\nabla h_2} - \mu_3 \underbrace{\begin{pmatrix} 0 \\ -1 \end{pmatrix}}_{\nabla h_3} = 0 \quad (10)$$

(There are no equality constraints.)

$$h_1(x) = x_1 + x_2 - 2 \leq 0 \quad (11)$$

$$h_2(x) = -2x_1 + x_2 + 1 \leq 0 \quad (12)$$

$$h_3(x) = -x_2 \leq 0 \quad (13)$$

$$\mu_i h_i(x) = 0 \quad \forall i \quad (14)$$

$$\mu \geq 0. \quad (15)$$

For a minimization problem only the lagrangian changes. Instead of $\nabla f - \sum \mu_i \nabla h_i$ we have $\nabla f + \sum \mu_i \nabla h_i$.

b) We investigate whether $x = (1, 1)$ is a local maximizer. Since $x_2 > 0$ it follows by complementary slackness 14 that $\mu_3 = 0$. Inserting $x_1 = x_2 = 1$ into 10 yields

$$\begin{aligned} \cos(1) - \mu_1 + 2\mu_2 &= 0 \Leftrightarrow \mu_1 = \cos(1) + 2\mu_2 \\ -\sin(1) - \mu_1 - \mu_2 &= 0. \end{aligned}$$

By substituting we get

$$-\sin(1) - \cos(1) - 2\mu_2 - \mu_2 = 0 \Leftrightarrow \mu_2 = -\frac{\sin(1) + \cos(1)}{3} < 0.$$

However, this contradicts condition 15. Thus, the KKT conditions do not hold for $(1, 1)$, so this point cannot be a local maximizer. In summary, if the KKT-conditions are necessary and a point is not a KKT-point, it can not be a local optimum.

c) Because $\mu_i = 0 \quad \forall i$ it follows from 10 that we have to solve $\nabla f(x) = 0$. This is equivalent to $\cos(x_1) = 0$ and $-\sin(x_2) = 0$. Thus, we obtain $x_{1,k} = \pi k - \frac{\pi}{2}$ for $k \in \mathbb{Z}$

and $x_{2,n} = \pi n$ with $n \in \mathbb{Z}$.

We have to find k and n such that $(x_{1,k}, x_{2,n})$ is feasible.

From $x_1, x_2 \geq 0$ we infer that

$$\begin{aligned} x_1 \geq 0 &\Leftrightarrow \pi k \geq \frac{\pi}{2} \Leftrightarrow k \geq \frac{1}{2} \\ x_2 \geq 0 &\Leftrightarrow n \geq 0. \end{aligned}$$

Condition 11 implies that

$$x_1 + x_2 = \pi k - \frac{\pi}{2} + \pi n = \pi(k + n) - \frac{\pi}{2} \leq 2 \Leftrightarrow k + n \leq \frac{2}{\pi} + \frac{1}{2} \approx 1.13.$$

Thus we receive $k = 1, n = 0$ as the unique solution. Consequently, it holds that $x_1 = \pi/2, x_2 = 0$. Furthermore we can check directly that all further KKT conditions hold for this point. Thus, we have computed the KKT-point corresponding to the multiplier $\mu = 0$.

Since the KKT conditions are necessary, but in general not sufficient for a loc. optimum, this is not sufficient to show local optimality of $x^* = (\pi/2, 0)$.

We have to use a different approach to show local optimality. In this case, we can even show that x^* globally maximizes the function f on $\mathbb{R}^2$, that is, without the constraints. To do this, we compute $f(x^*) = \sin(\pi/2) + \cos(0) = 1 + 1 = 2$. Obviously, it holds for any x_1, x_2 that $\sin(x_1) + \cos(x_2) \leq 2$. Therefore x^* is a global optimum on $\mathbb{R}^2$, so in particular also on the admissible domain. Since the set of global maxima is given by $x_1 = \pi/2 + 2k\pi$ and $x_2 = 2l\pi$, it follows that x^* is the only global maximum in the admissible domain.

- d) The KKT conditions are necessary for optimality. Furthermore, since the feasible region is compact, the continuous function f admits a global minimizer. Thus, there exists a global minimizer, which has to be a KKT-point.

In c) we computed a KKT-point with multiplier $\mu = 0$. According to a) this is a KKT-point for minimization as well. Furthermore, we know that there is only one KKT-point $(\frac{\pi}{2}, 0)$ with $\mu = 0$. Consequently, all other possible KKT-points lie on the boundary of the feasible region.

One could compute all KKT-points on the boundary and compare their objective value. However, we present an alternative that works faster in this case, since the feasible set is a Polytope and thus the boundary has a simple form. We compare the objective value of all points on the boundary to the objective value of $f(1, 1) = \sin(1) + \cos(1) \approx 1.38$.

We use that a strictly concave twice continuously differentiable function $f : [a, b] \rightarrow \mathbb{R}$ admits its minimum only at the endpoints a or b . We show that the objective restricted to each boundary segment is strictly concave. Thus we only need to compare the objective values corresponding to the corners of the Polytope (C_1, C_2, C_3) .

$$\begin{array}{c} C_3 = (1, 1) \\ \text{II} \quad \triangle \quad \text{III} \\ C_1 = (0.5, 0) \quad \text{I} \quad C_2 = (2, 0) \end{array}$$

- I) $y = 0, x \in [0.5, 2]$:

$$f(x, 0) = \sin(x) + \cos(0) = \sin(x) + 1$$

$\frac{d^2}{dx^2} \sin(x) = -\sin(x) < 0$ for $x \in [0.5, 2]$. Thus, $f(x, 0)$ is strictly concave on the interval $[0.5, 2]$.

II) $x \in [0.5, 1]$, $y = 2x - 1$:

$$f(x, 2x - 1) = \sin(x) + \cos(2x - 1)$$

$$\frac{d^2}{dx^2}(\sin(x) + \cos(2x - 1)) = -\sin(x) + -4\cos(2x - 1) < 0 \text{ for } x \in [0.5, 1].$$

III) $x \in [1, 2]$, $y = 2 - x$: $f(x, 2 - x) = \sin(x) + \cos(2 - x)$

$$\frac{d^2}{dx^2}(\sin(x) + \cos(2 - x)) = -\sin(x) + -\cos(2 - x) < 0 \text{ for } x \in [1, 2].$$

$$f(C_1) = f(0.5, 0) \approx 1.48$$

$$f(C_2) = f(2, 0) \approx 1.91$$

$$f(C_3) = f(1, 1) \approx 1.38$$

As an alternative the KKT-points and their respective multipliers are given in the table below.

μ_1	μ_2	μ_3	x	y	f	f rounded
0	0	0	$\frac{\pi}{2}$	0	2	2.0
0	$\frac{\cos(\frac{1}{2})}{2}$	$\frac{\cos(\frac{1}{2})}{2}$	$\frac{1}{2}$	0	$\sin(\frac{1}{2}) + 1$	1.48
$-\frac{\cos(1)}{3} + \frac{2\sin(1)}{3}$	$\frac{\sqrt{2}\sin(\frac{\pi}{4}+1)}{3}$	0	1	1	$\cos(1) + \sin(1)$	1.38
$-\cos(2)$	0	$-\cos(2)$	2	0	$\sin(2) + 1$	1.91

Table 1: KKT-points for minimization